MINING DEALER BEST PRACTICE

Repositioning of the Connection Box - CAT 7495

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

December 2022
1.01-221212-1

1.0 Introduction

The purpose of this Best Practice is to facilitate inspections on the **PN 514-3177** speed sensors of the two (02) trolley engines of 7495 Rope Shovel. On **Caterpillar 7495** Robe Shovels, the trolley motor gear has come into contact with one of the faces of the speed sensor causing it to fail.

With guidance from Caterpillar, through the Service request profile **(CRM-1710616-C1B0)**. The Sotreq Team inserted shims on the base of the locomotion stick to regulate the speed reading distance and avoid physical contact with the locomotion gear.

The technicians make inspections to verify the physical conditions of the club, to verify that it is not in contact with the gear of the locomotion engine. However, for the inspection of the locomotion stick, it is necessary to remove the **Box AS PN 466-0445** to access the locomotion sensor **PN 514-3177**, this action takes time, which can impact the Physical Availability of the equipment. It is worth noting that this **Box AS** works as a protection against material impacts on the locomotion sensor.

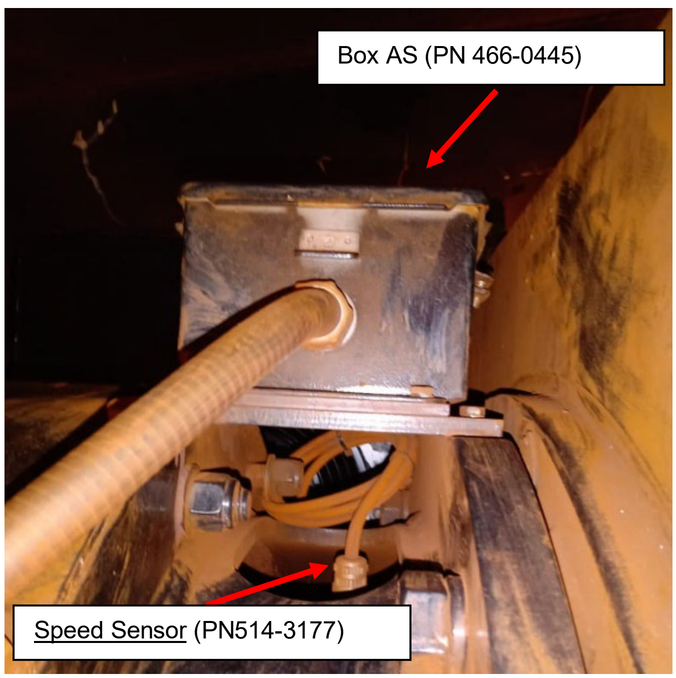

Figure 1: Location of "**Box AS**" **PN 466-0445** and "Speed Sensor" **PN 514-3177** on the Locomotion Engine.

To access the speed sensor, it is necessary to remove the **PN 460-9444** plate (06), **PN 1H-3337** screws (16), **PN 3B-4504** washers, and **PN 466-0444** plate (30). See the pic below:

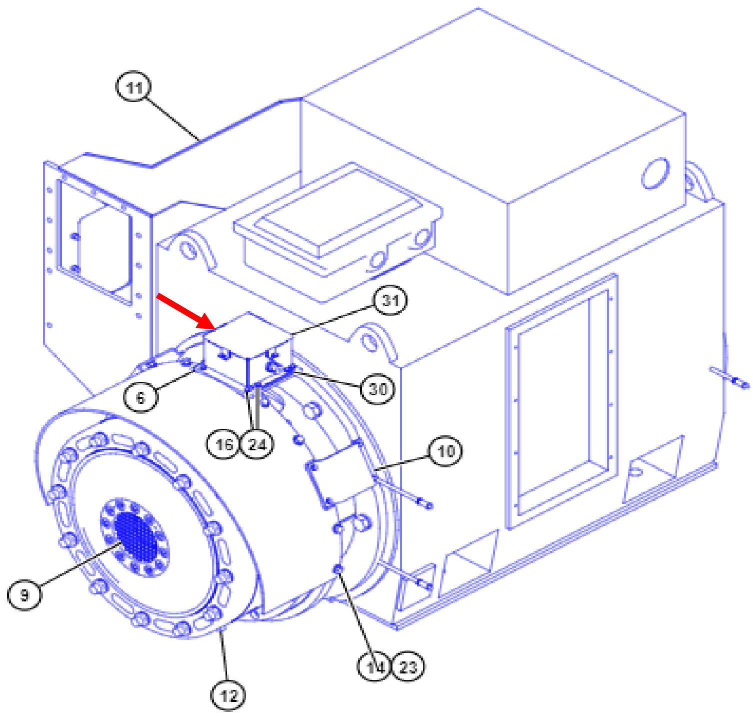

Figure 2: Location of "**Box AS**" **PN 466-0445** on the Locomotion Engine.

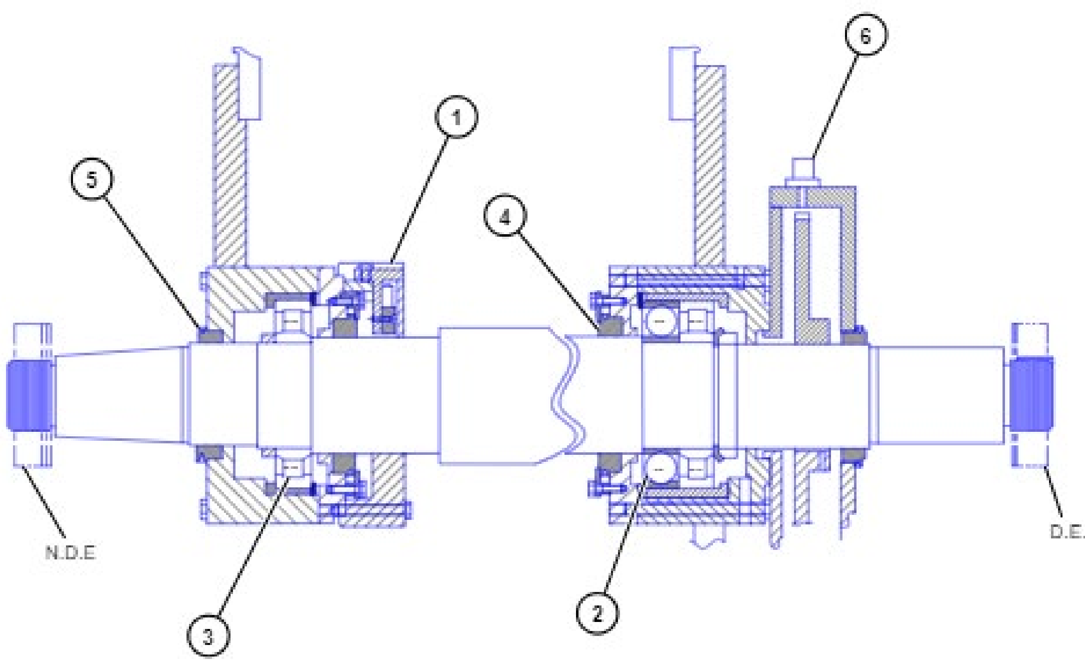

Figure 3: Location of the "Speed Sensor" **PN 514-3177** on the Locomotion Engine.

2.0 Best Practice Description

This Best Practice facilitates access to the Locomotion Engine Speed Sensor, protects against rock impact, and reduces maintenance and inspection time, through a hinged cover positioned above the locomotion stick.

This Best Practice consists of repositioning the **Box AS** from point (A) to point (B) and installing a plate with joints in the place where the Box was previously positioned.

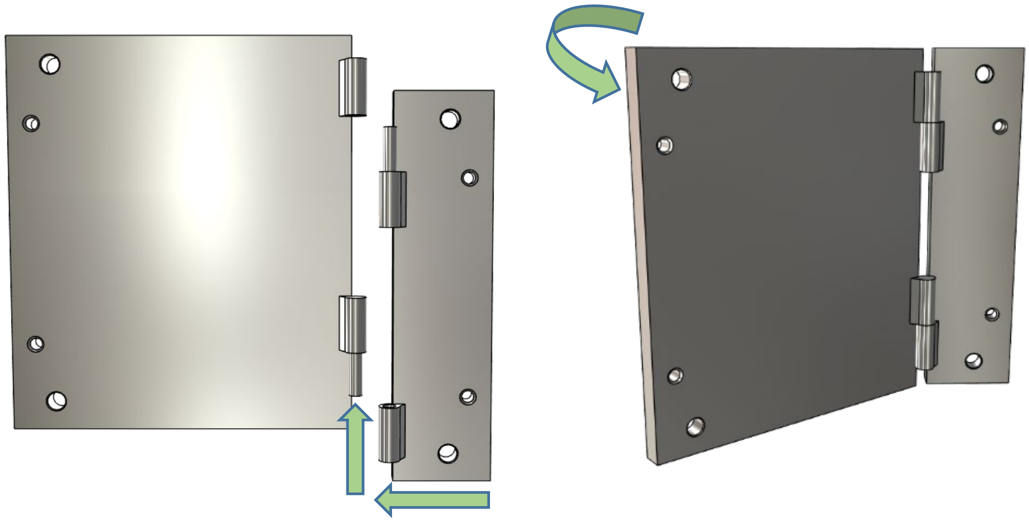

Figure 4 – 5: Perspective Demonstration of the hinged plate.

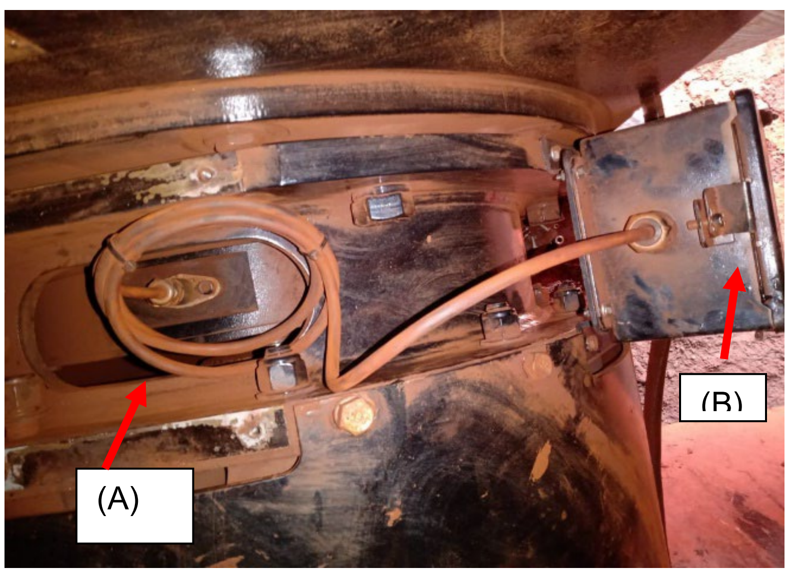

Figure 6: Point (A) and Point (B).

3.0 Implementation Steps

To implement this **Best Practice**, follow the steps below:

1- Remove Bolts **PN 1H-3337** from the Box fixing base.
2- Remove Washers **PN 3B-4504** from the housing fixing base.
3- Remove Plates **PN 466-044** and **460-9444**.
4- Remove Box **PN 466-0445.**
5- Fix the articulated Plate specified in Figure 7.
6- Fix the Plate with the **PN 1H-3337** Bolts.
7- Reposition Box **PN 466-0445** in position (B) indicated in Figure 6.
8- Fix the Box with the same set mentioned to remove it.

All the measurements are determined for the manufacture of the hinged plate. The plate was made of **ASTM A36** steel. See the pic below:

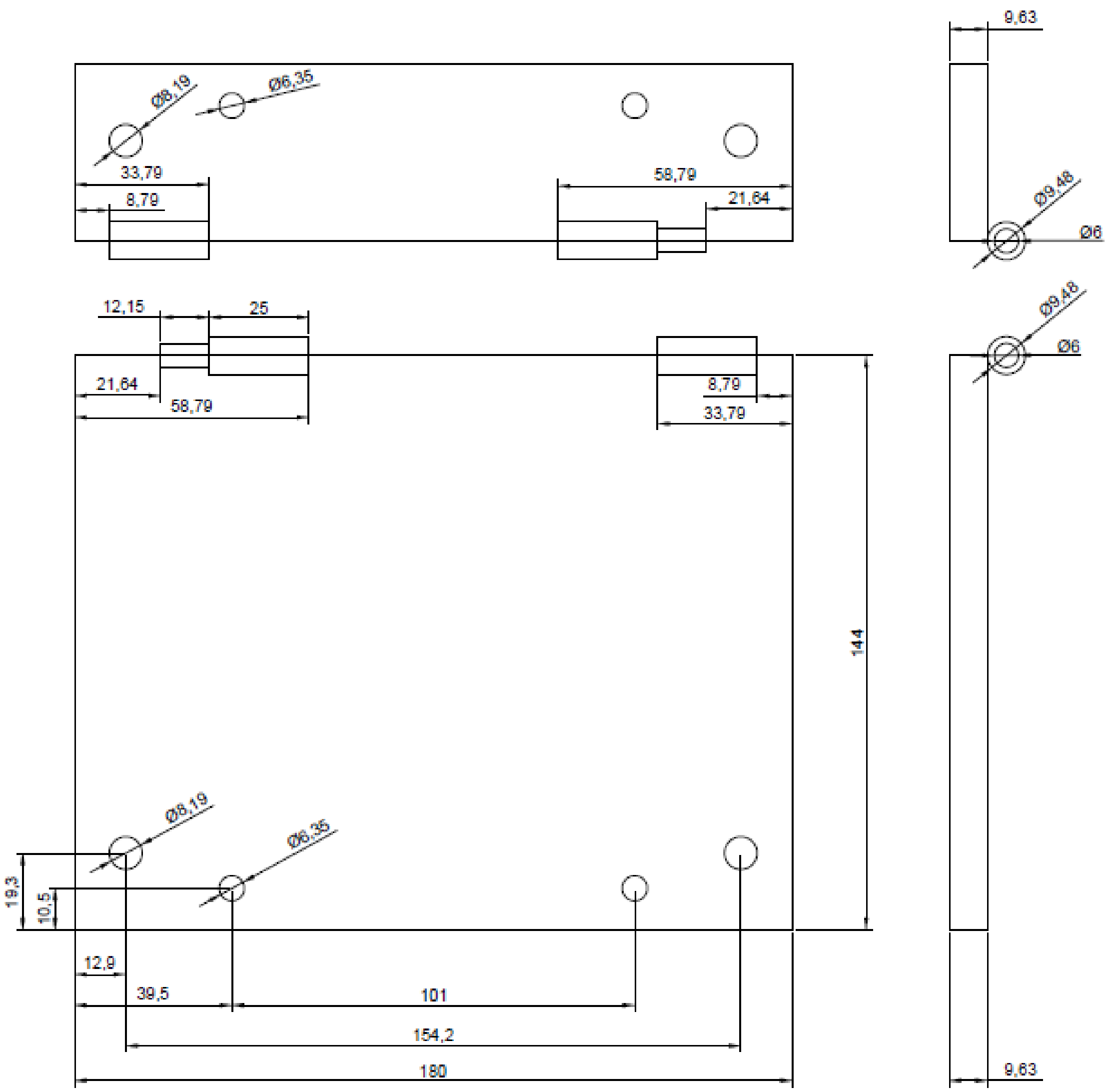

Figure 7: 3D Drawing of the plate.

For the installation of the plate, the same adjustments of the base of **PN 1H-337** were used.

Figure 8: Plate Fixing.

4.0 Benefits

- Decreased speed sensor inspection and maintenance time by 60% of the time.
- Facilitates the visual inspection process during maintenance.
- Improved accessibility.

5.0 Resources Required

The software "**AutoCAD**" (Autodesk) was used for the **3D** drawing of the plate.

In the development of this project, original Caterpillar parts were used as a way to guarantee the reliability and effectiveness of the project. The quantities presented below were calculated for application in the two locomotion engines.

Item	Qty (Unity/Length)	PN
Speed Sensor	02	514-3177
Plate	02	460-9444
Bolt	16	1H-3337
Lockwasher	32	3B-4504

6.0 Supporting Attachments / References

Figure 03 - https://sis2.cat.com/#/detail?keyword=460-9438&serialNumber=ER6&ieSystemControlNumber=NIS000731075&componentId=00002201&tab=parts.

Figure 04- https://sis2.cat.com/#/detail?keyword=514-3177&tab=parts&serialNumber=ER6&ieSystemControlNumber=NIS000701972&componentId=0000032.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

I thank all the effort and dedication of the engineering team of the expanded mining Products of the dealer Sotreq.

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